



IE 4197: System Analysis and Design

Lecture 2a: Analysing Audience and Purpose



Analysing Your Audience and Purpose



- ▶ Technical communication does not happen in vacuum
- ▶ Audience: Who will be reading your document
- ▶ Purpose: Why will they read your document



Who Are Your Readers?



- ▶ Education
- ▶ Professional experience
- ▶ Job responsibility
- ▶ Personal characteristics and preferences
- ▶ Cultural characteristics



What Will Your Audience Do?



- ▶ File it?
- ▶ Skim it?
- ▶ Read some of it?
- ▶ Read the whole?
- ▶ Study it?
- ▶ Modify it?



Who Else Will Read Your Document?



- ▶ Primary audience: People who are supposed to act on the correspondance
- ▶ Secondary audience: People who should be informed about the contents of the correspondance
- ▶ Tertiary audience: Other groups or organizations who might be interested



Communicating Across Cultures



- ▶ Cultural variables on the surface:
 - Political
 - Economic
 - Social
 - Religious
 - Educational
 - Technological
 - Linguistic



Communicating Across Cultures



- ▶ Some cultures value individuals more than groups
 - Use “I” rather than “we”
 - Don’t use formal titles very often
- ▶ Some cultures put more distance between business and private life
 - Information about personal life is seldom included in professional correspondence



Communicating Across Cultures



- ▶ Some cultures put less distance between ranks
 - Communication is less formal
 - Inappropriate informality must be avoided
- ▶ Some cultures feel that truth is a universal concept
 - Documents are detailed and comprehensive
 - Conclusions are reached, not left to readers' interpretation



Communicating Across Cultures



- ▶ In some cultures it is necessary to spell out all the details
 - Documents and contracts are lengthy and detailed (low context culture)
 - On the other hand, writing obvious details is considered an insult to readers' intelligence in some cultures (high context culture)
- ▶ In some cultures people are not comfortable with uncertainty
 - Communication is usually written
 - Policies are lengthy and specific



Guidelines for Comm. Across Cultures



- ▶ Limit your Vocabulary
- ▶ Keep sentences short
- ▶ Define abbreviations and acronyms
- ▶ Avoid jargon, idioms and slang
- ▶ Use active voice
- ▶ Be careful with graphics
- ▶ If possible, have someone from the culture review your document



Purpose



- ▶ Communicating ideas or information
 - Describe
 - Illustrate
 - Explain
 - Summarize
- ▶ Convincing an opinion
 - Assess
 - Recommend
 - Propose
 - Request



Reference



- ▶ Markel, M. (2010) *Technical Communication (9th ed.)*. Boston, MA: Bedford/St. Martin's



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Lecture 2b: Writing Effective Sentences



Outline



- ▶ What are the characteristics of effective sentences?
- ▶ How to structure effective sentences?
- ▶ How to chose words for effective sentences?



Effective Sentences



- ▶ Accurate
- ▶ Clear
- ▶ Concise
- ▶ Easy to understand
- ▶ Emphasis on what is being said, not how it is said



Ineffective Sentences



- ▶ May cause misunderstandings.
- ▶ May not be understood at all.
- ▶ May cause costly problems.
- ▶ Reflects poorly on the writer and organization.



Structure of Effective Sentences



- ▶ Use lists.
- ▶ Emphasize new and important information.
- ▶ Use sentences with appropriate length.
- ▶ Focus on the real subject.
- ▶ Focus on the real verb.
- ▶ Use parallel structure.
- ▶ Use proper modifiers.



Lists



- ▶ Easy to remember and understand.
- ▶ Break up long lists with 10 or more items.
- ▶ Use a parallel structure.
- ▶ Use appropriate lead-in.
- ▶ Use correct punctuation.



Emphasizing New and Important Information



- ▶ Place the new information to the end of the sentence.
Because of labor problems, we anticipate a three-week delay in production
- ▶ Don't end the sentence with qualifying information
~~*The joint could fail under special circumstances.*~~
Under special circumstances, the joint could fail.
- ▶ Put orienters to time and space to the beginning of the sentence
On the north side of the building, water from the leaking pipes has damaged the exterior siding and the drywall on some of the interior walls.



Appropriate Sentence Length



- ▶ 15–20 words on average is effective for most technical writing.
- ▶ Long sentences are confusing, short sentences are sluggish and choppy, a series of sentences with the same length is monotonous.
- ▶ Long sentences can be broken down to lists.

Focusing on the Real Subject

- ▶ The real subject of the sentence should also be its grammatical subject.

- ▶ Prepositional phrases weaken the subject.

~~*The use of this method would eliminate the problem of motor damage.*~~

This method would eliminate the problem of motor damage.

- ▶ Avoid expletives if possible

~~*There is no alternative for us except to withdraw the product.*~~

We have no alternative except to withdraw the product.



Focusing on the Real Verb



- ▶ Avoid nominalized verbs coupled with weak verbs

~~*Installation of the software was performed.*~~

Software was installed.

~~*Data analysis is done.*~~

Data is analyzed.

- ▶ Nominalizations are effective as subjects.

Using Parallel Structure

- ▶ Elements in a sentence should follow the same grammatical form

~~*Our present system is costing us profits and reduces our productivity.*~~

Our present system is costing us profits and reducing our productivity.

- ▶ Avoid overlapping or otherwise confusing parallel items

~~*The speakers will include partners of the law firms, businesspeople and civic leaders.*~~

The speakers will include business people, civic leaders and partners of the law firms.

- ▶ Words, phrases and clauses that describe other elements in the sentence are called modifiers.
- ▶ Restrictive modifier: modifier required to identify the referent.

Please disregard the notice you recently received from us.

- ▶ Nonrestrictive modifier: modifier that provides additional information.

When you arrive, go to the Registration Area, which is located on the second floor.

- ▶ Use commas to separate a nonrestrictive modifier from the rest of the sentence.

Using Modifiers Appropriately

- ▶ Avoid misplaced modifiers.

~~*The subject of the meeting is the future of geothermal energy in the downtown Webster Hotel.*~~

The subject of the meeting in the downtown Webster Hotel is the future of geothermal Energy.

- ▶ Avoid dangling modifiers.

~~*Trying to solve the problem, the instructions seemed unclear.*~~

As I was trying to solve the problem, the instructions seemed unclear.



Elements of Effective Sentences



- ▶ Select appropriate level of formality
- ▶ Use appropriate voice
- ▶ Avoid ambiguity
- ▶ Use positive constructions
- ▶ Avoid long noun strings
- ▶ Be concise

- ▶ Although there is no universal standard, it is possible to classify writing into three categories:
 - Informal: “The Acorn 560 is a real screamer. With 3.8 GHz of pure computing power, it slashes through even the thickest spreadsheets before you can say $2 + 2 = 4$.”
 - Moderately formal: “With its 3.8 GHz microprocessor, the Acorn 560 can handle even the most complicated spreadsheets quickly.”
 - Highly formal: “With a 3.8 GHz microprocessor, the Acorn 560 is a high-speed personal-computer appropriate for the computation-intensive applications such as large, complex spreadsheets.”



Level of Formality



- ▶ Technical writing is moderately to highly formal
- ▶ Level of formality required depends on:
 - Audience
 - Subject
 - Purpose
- ▶ It is better to err on the side of higher formality.

Active and Passive Voice

- ▶ Active voice puts emphasis on the subject that performs the verb
- ▶ Passive voice puts emphasis on the subject that is the recipient of the action
- ▶ Active voice is more appropriate, except:
 - When the agent is clear from the context
 - When the agent is unknown
 - When the agent is less important than the action
 - When a reference is embarrassing, dangerous or inappropriate

Avoiding Ambiguity

- ▶ Use precise words
- ▶ Provide adequate detail
 - ~~*An engine of the plane experienced some difficulties.*~~
~~*The left engine on the Martin 411 temporarily lost power during flight.*~~
- ▶ Avoid expressions that can have multiple meanings
 - ~~*After stirring by hand for 10 seconds, add three drops of the iodine mixture to the solution.*~~

- ▶ Tell the reader what something is.
- ▶ Positive constructs are easier and quicker to understand.
- ▶ Use “most”, “few”, “inefficient”, “on time”; instead of “not all”, “not many”, “not efficient” and “not delayed”.

~~*Because the team did not have sufficient time to complete the project, it was not surprising that it was not able to prepare a satisfactory report.*~~

Because the team had too little time to complete the project, it produced an unsatisfactory report.

Noun Strings

- ▶ Strings of nouns (as well as adjectives and adverbs) modifying the succeeding one.
- ▶ Long noun strings lack clarity
~~preregistration procedures instruction sheet update~~
~~update of the instruction sheet for preregistration~~
~~procedures~~
- ▶ Some noun strings may be ambiguous
~~The building contains a special incoming materials~~
~~storage area.~~
The building contains a special area for storing
incoming materials.
The building contains an area for storing special
incoming materials.

- ▶ Avoid obvious statements.

The market for ~~the sale of~~ flash memory chips is dominated by ~~two chip manufacturers~~: Intel and Advanced Micro Systems. ~~These two chip manufacturers~~ are responsible for 76 percent of the \$1.3 billion market ~~in the flash memory chips last year~~.

- ▶ Avoid unnecessary filler such as “basically”, “certain”, “sort of”. They are more appropriate in oral communication.

- ▶ Avoid unnecessary prepositional phrases.
The increase in the number of students enrolled in the materials engineering program at Lehigh University is suggestive of the regard in which that program is held by the university's new students.
The growth of Lehigh University's materials engineering program suggests that the university's new students consider it a good program.
- ▶ Avoid wordy phrases. Use “most”, “now”, “can”, “daily” instead of “the majority of”, “at the present time”, “have the capability to”, “on a daily basis”.



Preparing Text for Translation



- ▶ Many documents are translated to other languages
- ▶ In order to make translations easier
 - Use short sentences (<20 words on average)
 - Use active voice
 - Use simpler words
 - Include a glossary in technical documents
 - Avoid words with more than one meaning
 - Avoid jokes, puns and culture-bound references



Reference



- ▶ Markel, M. (2010) *Technical Communication (9th ed.)*. Boston, MA: Bedford/St. Martin's



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Lecture 2c: Work packages



Work Packages: General Definition



- ▶ Efficient project management requires a work hierarchy (WBS)
- ▶ Lowest level of WBS is called Work Package
- ▶ Work Packages:
 - Contain actual work (activities) to be done
 - Have start and end dates
 - Consume resources



Work Packages in IE4197



- ▶ 4 Work Packages for IE4197 (4 more for 4198):
 - Problem Definition
 - Observation and Literature Survey
 - Theoretical model and Methods
 - Data verification and Tools
- ▶ One student is responsible for each WP
- ▶ WPs must be documented on the report



WP1 : Problem Definition



- ▶ September 29th – October 17th
- ▶ Initial observations about the problem
- ▶ Reasoning about possible solutions
- ▶ Determination of the project scope
- ▶ Deliverable: Project Approval Form



WP2: Observation and Literature Survey



- ▶ October 20th – November 7th 2025
- ▶ Examination of prior solutions to similar problems
- ▶ Evaluation of possible methods/tools/algorithms
- ▶ Determination of data requirements
- ▶ **Interim Report due: November 19th**



WP3: Theoretical model and Methods



- ▶ November 24th –December 12th 2025
- ▶ Model proposal based on the evaluation of possible models
- ▶ Components of the proposed solution
- ▶ Data Collection
- ▶ Deliverable: Theoretical Model



WP4: Data verification and Tools



- ▶ December 15th 2025– January 2nd 2026
- ▶ Data cleaning/verification
- ▶ Data analysis
- ▶ Determination of tools to be used
- ▶ Scenarios, parameters, criteria
- ▶ Deliverable: Data, scenarios, methods



Week 15: Deliverables



- ▶ January 5th : Reports due
- ▶ January 7th : Final Presentations



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Lecture 3a: Creating Graphics



Purpose of Graphics



- ▶ Catching attention
- ▶ Communicating information that is difficult to communicate with words
- ▶ Clarify and emphasize information
- ▶ Help nonnative speakers



Functions of Graphics



- ▶ Demonstrating logical and numerical relationships
- ▶ Communicating spacial information
- ▶ Communicating steps in a process
- ▶ Saving space
- ▶ Reducing cost of translations



Characteristics of an Effective Graphic



- ▶ Serves a purpose
- ▶ Is simple and uncluttered
- ▶ Presents manageable amount of information
- ▶ Is clearly labeled
- ▶ Meets readers' format expectations

Types of Graphics: Table

- Best for presenting large amounts of numeric data
- Arrange items being compared in rows
- Include units of measure
- Minimize use of rules

2009-2010 Grain Production in Turkey

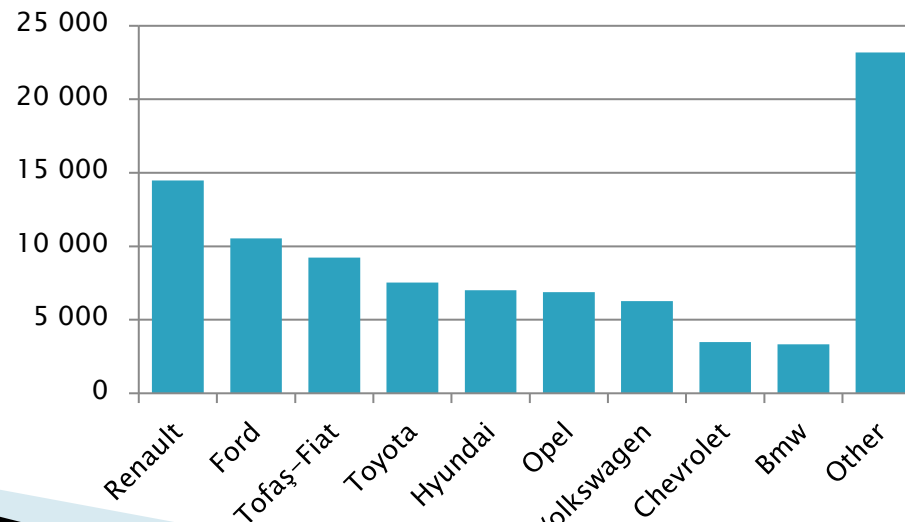
Grain	2009 Production		2010 Production		Change
	(Tons)	Share (%)	(Tons)	Share (%)	
Wheat	20 600 000	61.56	19 660 000	60.23	-4.6
Maize	4 250 000	12.70	4 310 000	13.20	1.4
Rice	750 000	2.24	860 000	2.63	14.7
Barley	7 300 000	21.82	7 240 000	22.18	-0.8
Rye	343 330	1.03	365 560	1.12	6.5
Oat	218 286	0.65	203 870	0.62	-6.6
Total	33 461 616	100	32 639 430	100	-2.5

Source: www.tuik.gov.tr

Types of Graphics: Bar graph

- Best for showing relative values of several items
- Vertical axis should be about 25% shorter than horizontal axis
- All bars should have same width, gap between bars should be about half as wide as a bar (exception: Histograms)
- Origin should be placed at zero if possible. Breaks should be clearly marked
- Bars should be arranged in a logical sequence

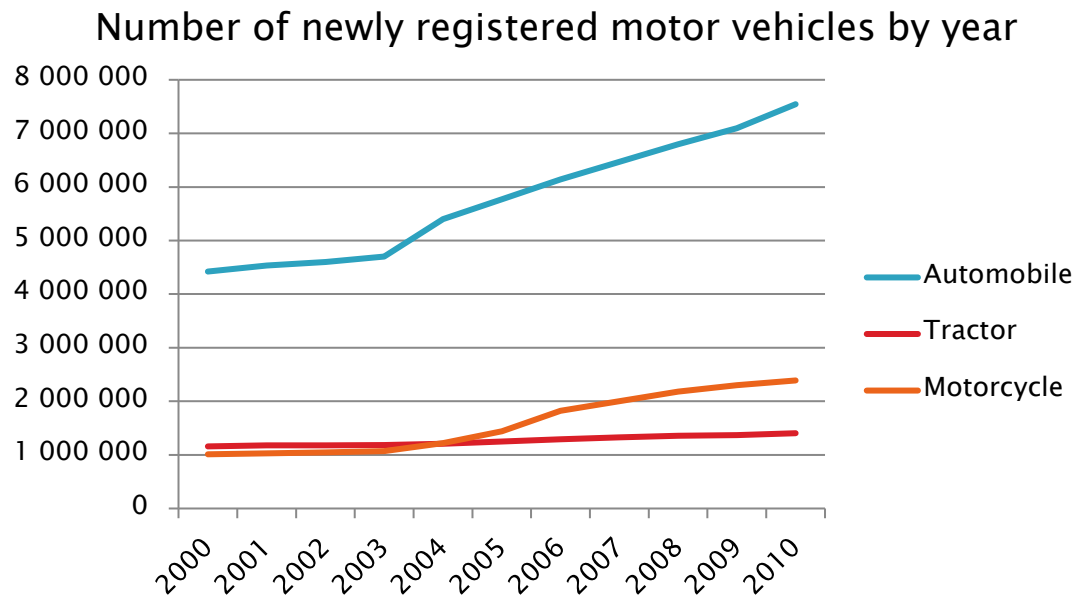
Breakdown of newly registered passenger vehicles in January 2011 by brands



Source: www.tuik.gov.tr

Types of Graphics: Line Graph

- Best for showing change of quantity over time
- Don't use more than 4 lines
- Put the origin at zero if possible. Mark the break clearly, if it is necessary.

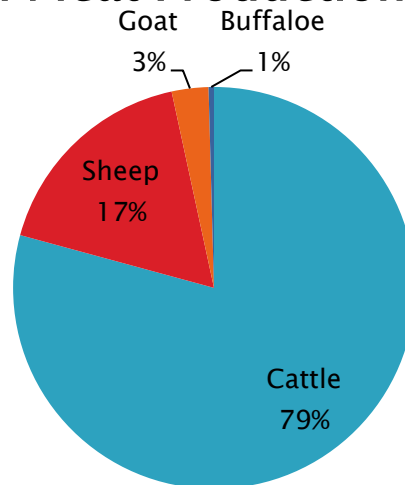


Source: www.tuik.gov.tr

Types of Graphics: Pie Chart

- Best for showing relative sizes of parts of a whole
- Don't use more than 7 slices
- Don't use 3D pies
- Label the slices horizontally, and in the slice if possible
- A slice can be emphasized by separating from the rest

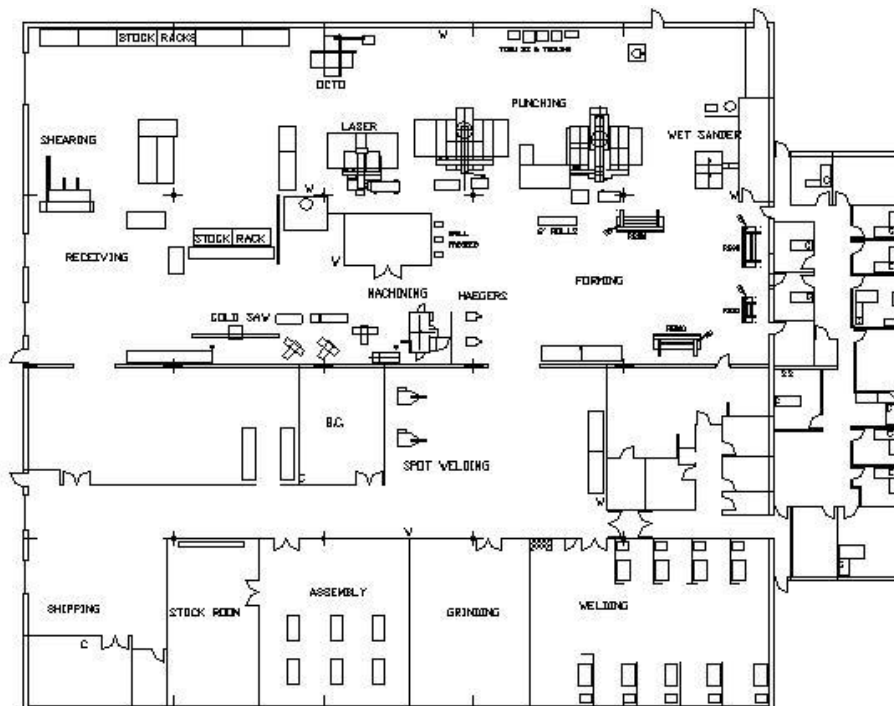
2010 Red Meat Production in Turkey



Source: www.tuik.gov.tr

Types of Graphics: Diagram

Uses symbols to represent items and their properties e.g. blueprints, wiring diagrams, schematics

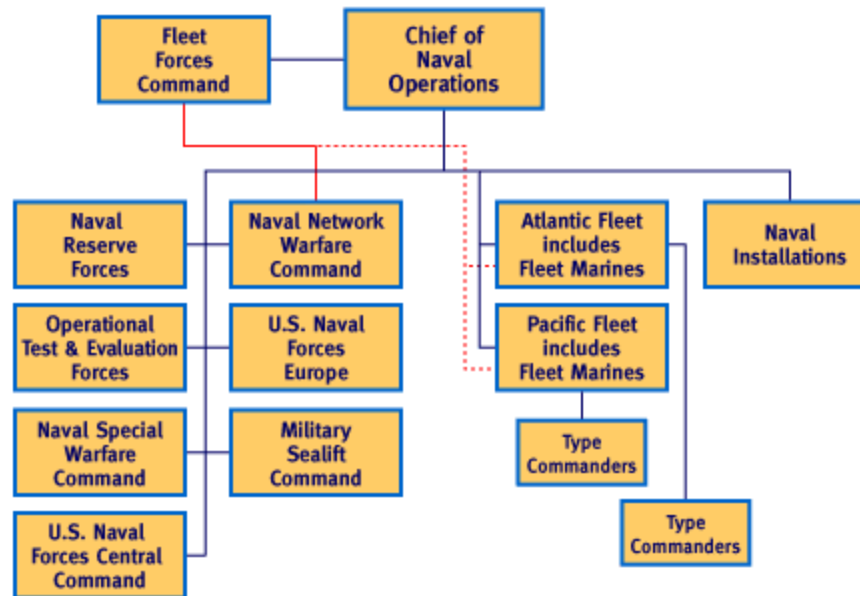


Source: www.ssmnh.com

Types of Graphics: Organizational chart

Uses simple geometric shapes to present logical or organizational relationships

US Navy Operating Forces Organizational Chart

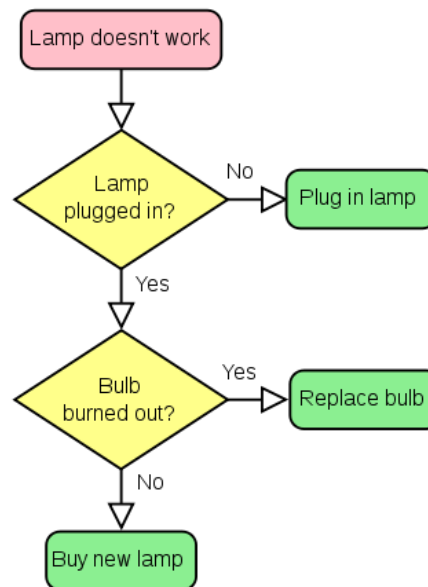


Source: www.navy.mil

Types of Graphics: Flowchart

- Shows the stages of a process or procedure
- Open flowcharts have a distinct start and finish

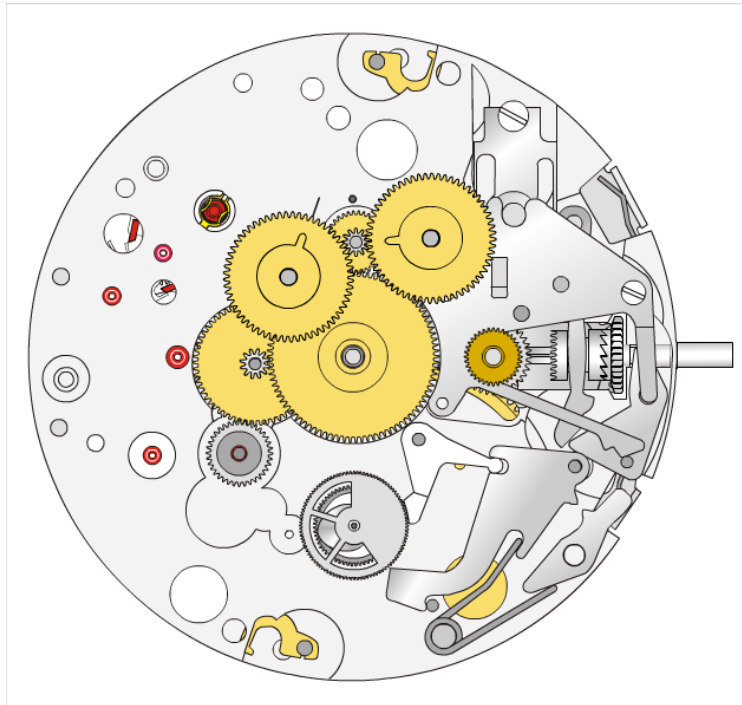
Lamp Flowchart



Source: www.wikipedia.org

Types of Graphics: Drawing

- Simplified visual representation of objects
- Focuses readers' attention on desired information



Source: www.eta.ch

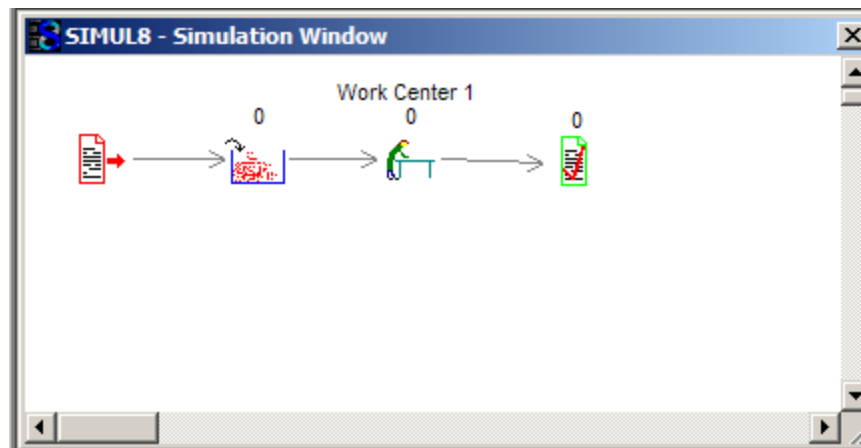
Types of Graphics: Photograph

Good for presenting large amount of detail



Types of Graphics: Screenshot

- Shows the reader what the computer screen looks like when the program is used
- Cursor should point to an area that is emphasized or not be visible



Source: M.C. Demir



Graphics and Ethical Considerations



- ▶ If you did not generate the graphic or data, cite your source. Obtain permission if necessary.
- ▶ Include all relevant data.
- ▶ Begin the axes of graphs at zero (or mark breaks clearly).
- ▶ Do not use a table to hide a data point that would be obvious in a graph.
- ▶ Show items as they are, do not manipulate photos or screenshots.
- ▶ Do not use color or shading to misrepresent an item's importance e.g. light shaded bar in a bar graph seems larger and nearer.



Integrating Graphics and Text



- ▶ Place graphics in an appropriate location.
 - If the graphic is needed for the reader to understand the discussion, put it as soon as possible after the discussion.
 - If the graphic merely supports or elaborates your point, put it in an appendix.
- ▶ Introduce the graphic in text.
 - Refer them by a number, e.g. “figure 14” or “appendix A”
 - Don’t use expressions like “figure above” or “table on the next page”.
- ▶ Explain graphic in the text.

- ▶ Demir, M.C. (2005) *Simul8labs*
- ▶ ETA (2011) *Disassembly of ETA 7750*. Accessed electronically at www.eta.ch/swisslab/7750/7750.html on March 29th, 2011
- ▶ Hubblesite (2011) *Helix Nebula*. Accessed electronically at www.hubblesite.org on March 24th, 2011
- ▶ Markel, M. (2010) *Technical Communication (9th ed.)*. Boston, MA: Bedford/St. Martin's
- ▶ US Navy (2011) *The Operating Forces*. Accessed electronically at www.navy.mil/navydata/organization/orgopfor.asp on March 24th, 2011
- ▶ Sylvester Sheet Metal Corporation (2011) *Facility Layout*. Accessed electronically at <http://www.ssmnh.com/facility.htm> on March 24th, 2011
- ▶ TUIK, (2011), *News Bulletins*. Accessed electronically at www.tuik.gov.tr on March 24th, 2011
- ▶ Wikipedia (2011) *Flowchart*. Accessed electronically at <http://en.wikipedia.org/wiki/Flowchart> on March 24th, 2011



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Lecture 3b: Making Oral Presentations



Types of Presentations



- ▶ **Impromptu presentations:**
Delivered without advance notice or preparation
- ▶ **Extemporaneous presentations:**
Presentation is planned and rehearsed, sentences are created on the go
- ▶ **Scripted presentations:**
Prepared text is read
- ▶ **Memorized presentations:**
Memorized speech is delivered without notes



Different Media Used in Presentations



- ▶ Computer presentation software:
 - Versatile
 - Professional appearance
- ▶ Audiovisuals:
 - Usually embedded into presentation software
 - Make sure that the venue can accommodate
- ▶ Flipcharts:
 - Good for illustrating change over multiple steps
 - Cumbersome to change pages



Different Media Used in Presentations



- ▶ Writing board:
 - Cannot be used to create complicated graphs
 - Ineffective in large rooms
- ▶ Handouts:
 - Allows listeners to take notes
 - Not environmentally friendly
 - Distracts the attention of the audience
- ▶ Objects:
 - Distracts the attention of the audience
 - May not survive the presentation intact



Preparing a Presentation



- ▶ Analyze the speaking situation
 - What is your audience and purpose?
 - How much time do you have?
- ▶ Organize and develop the presentation
 - Develop a sensible organizational pattern e.g. problem–methods–solution
 - Have a clear main point (or a few main points)
 - Pay attention to introduction and conclusion



Preparing a Presentation



- ▶ Prepare graphics
 - They make presentation easier to remember
 - An effective graphic:
 - Presents clear, well-supported claim
 - Is easy to see and read
 - Is simple
 - Is correct
 - Use graphics that are appropriate for the time available and needs of the audience



Rehearsing a Presentation



Rehearse your presentation at least three times

- First rehearsal:
 - Is the presentation well organized?
 - Is the message clear?
 - Are the graphics appropriate?
- Second rehearsal:
 - Are you within the time budget?
- Third rehearsal:
 - Is the presentation satisfactory in every way?
 - Preferably in front of an audience



Delivering a Presentation



- ▶ Calm your nerves
 - The audience is most likely not aware of your nervousness
 - Nervousness gives you focus and energy
 - Nervousness will pass in a few minutes
- ▶ Introduce yourself
- ▶ State the title of your presentation
- ▶ Explain the purpose and main point of your presentation



Using Effective Language and Voice



- ▶ Pronunciation
- ▶ Volume
- ▶ Speed
- ▶ Advance organizers
 - “Now I’ll talk about...”
- ▶ Summaries
 - “We have seen that...”
- ▶ Transitions
 - “I’d like to turn to my second main point”



Facing the Audience



- ▶ Maintain eye contact
 - Let's you see how the audience is receiving
 - Don't stare at the same person
 - Don't look at a fixed point
- ▶ Use natural gestures
 - Gestures help reinforce the words
 - Avoid mannerisms
- ▶ Don't block the audience's view of the screen
 - Use a pointer if necessary



Answering Questions



- ▶ Invite questions
- ▶ Make sure:
 - Everyone has heard the question
 - You understood the question
- ▶ Don't insult the person
- ▶ Offer to discuss the issue privately to belligerent listeners



Reference



- ▶ Markel, M. (2010) *Technical Communication (9th ed.)*. Boston, MA: Bedford/St. Martin's
- ▶ <https://open.lib.umn.edu/publicspeaking/chapter/15-3-media-to-use-for-presentation-aids/>